Research Square

Preprints are preliminary reports that have not undergone peer review. They should not be considered conclusive, used to inform clinical practice, or referenced by the media as validated information.

Estimation of Battery Life Cycle & Remaining using Machine Learning

Ashish Khurana

ashish.khurana@s.amity.edu

Amity University

O P Singh

Amity University

Research Article

Keywords: RUL- Remaining Useful Life, ANN- Artificial Neural Networks, GBR- Gradient Boosting Regression, RFR- Random Forest Regression)

Posted Date: August 2nd, 2024

DOI: https://doi.org/10.21203/rs.3.rs-4595067/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License. Read Full License

Additional Declarations: No competing interests reported.

Estimation of Battery Life Cycle & Remaining using Machine Learning

[1]Ashish Khurana and [2] O P Singh
[1]*Department of Mechanical Engineering,*
[2]*Department of Electrical and Electronics Engineering*
Amity School of Engineering & Technology(ASET),
Amity University, Uttar Pradesh
Lucknow Campus, India
ashish.khurana@s.amity.edu, opsingh@amity.edu

***Abstract*—Accurately estimating the remaining useful life (RUL) and capacity of lithium-ion batteries essential for optimizing the performance, safety, and efficiency of various applications, including electric vehicles, portable electronics, and renewable energy systems. Traditional methods for predicting battery health often fall short due to the nonlinear and intricate degradation patterns of lithium-ion batteries. This research paper introduces an innovative machine learning-based approach to enhance the prediction accuracy of RUL and capacity for lithium-ion batteries. By leveraging comprehensive datasets from a custom-built battery prognostics testbed at the NASA Ames Prognostics Center of Excellence (PCoE), which includes real-time measurements of charge, discharge, and electrochemical impedance spectroscopy across different temperatures. This study aims to significantly improve battery health monitoring and prognostics**

Keywords*— *RUL- Remaining Useful Life, ANN- Artificial Neural Networks, GBR- Gradient Boosting Regression, RFR- Random Forest Regression)

I. INTRODUCTION

With the Lithium-ion batteries having revolutionized portable energy storage, powering devices ranging from smartphones and laptops to electric vehicles (EVs) and renewable energy systems. Their high energy density, lightweight construction, and long lifespan make them the preferred choice for numerous applications. However, the remaining useful life (RUL) of these batteries is a critical factor in determining their usability and economic viability. Accurate prediction of the RUL is essential for optimizing battery usage, ensuring safety, and planning timely replacements, impacting consumer satisfaction and the broader push towards sustainable energy solutions.

The RUL of a lithium-ion battery refers to the period during which it can effectively perform its intended function before degrading to an unusable state. This degradation is influenced by numerous factors, including battery chemistry, usage patterns, charging cycles, operating temperature, and storage conditions. As a battery ages, its capacity to hold charge diminishes, internal resistance increases, and the risk of failure or safety hazards, such as thermal runaway, escalates. Thus, predicting the RUL is not only a matter of convenience but also crucial for safety and reliability across applications from consumer electronics to large-scale energy storage systems.

Accurate RUL prediction involves a combination of empirical data collection, mathematical modeling, and advanced diagnostic techniques. Traditional approaches, such as cycle counting and capacity fade analysis, monitor battery performance over repeated charge-discharge cycles. More advanced methods utilize machine learning algorithms and artificial intelligence to analyze large datasets, identify patterns, and predict future performance based on historical data. Diagnostic tools like electrochemical impedance spectroscopy provide detailed insights into the battery's internal state, offering a deeper understanding of its health and longevity.

Understanding the RUL of lithium-ion batteries significantly contributes to environmental sustainability. Effective management of battery life cycles and timely replacements can reduce electronic waste and promote recycling and repurposing of battery components. This is particularly important as global demand for lithium-ion batteries rises, driven by the growth of electric mobility and renewable energy initiatives.

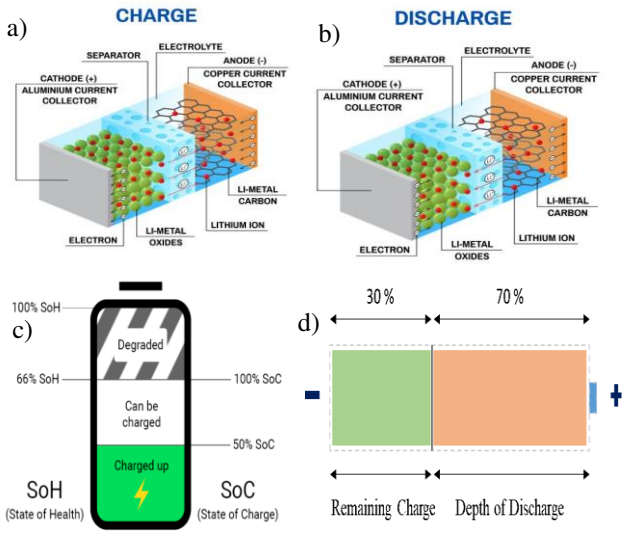


Fig. 1 a) Battery Charging Cycle b) Battery Discharging Cycle c) SOC d) Depth of Discharge (DoD)

In conclusion, the remaining useful life of lithium-ion batteries is a pivotal aspect that affects their performance, safety, and economic viability across various applications. As technology advances, methods for predicting and managing RUL are becoming increasingly sophisticated, promising enhanced reliability, efficiency, and sustainability in the use of these essential energy storage devices. Improving the understanding and management of the RUL of lithium-ion batteries is not only a technical challenge but also a crucial step towards a more sustainable and energy-efficient future.

II. BATTERY CAPACITY

As we use and recharge a lithium-ion battery, its ability to operate at peak capacity gradually diminishes. The number of

$$Battery\ Capacity\ (Ah) = Current\ (A) \times\ Dischg\ Time\ (h) \quad (1)$$

charge and discharge cycles a battery can endure before losing efficiency is known as its life cycle. The capacity of a lithium-ion battery is defined as the total amount of electric charge it can store and deliver to an external circuit under specified conditions. This capacity quantifies the battery's ability to provide energy over a given period and is typically expressed in ampere-hours (Ah) or mill ampere-hours (mAh). Capacity is directly related to the amount of active material within the battery and the efficiency of the electrochemical reactions occurring during charge and discharge cycles.

Key aspects of battery capacity include:

Nominal Capacity: This is the manufacturer-specified capacity of the battery, determined under standardized test conditions, such as a specified discharge rate, temperature, and cutoff voltage. It represents the theoretical maximum amount of charge the battery can hold.

Usable Capacity: This refers to the actual capacity available for use in practical applications, which can be affected by factors such as the rate of discharge, operating temperature, and aging of the battery. Usable capacity is often lower than the nominal capacity due to these real-world conditions.

Discharge Capacity: This is the total charge the battery can deliver when discharged at a constant current until it reaches the cutoff voltage. It is typically measured in Ah or mAh and is an important parameter for determining the battery's performance in specific applications.

Energy Capacity: Often expressed in watt-hours (Wh), this is the product of the battery's capacity (Ah) and its nominal voltage (V). This metric provides a measure of the total energy the battery can store and deliver.

Capacity Fade: This is the gradual reduction in a battery's capacity over time due to factors such as repeated charge-discharge cycles, chemical degradation, and operational stress. Capacity fade is a key indicator of battery aging and health.

A. Battery Charging Cycle

The operation of a lithium-ion battery is based on the movement of lithium ions from the anode to the cathode through an electrolyte, facilitated by an external circuit. This ion mobility is the fundamental process behind charging and discharging. During battery charging cycle as shown in Fig 1 (a), lithium ions travel from the cathode to the anode, where they are stored, while electrons move from the cathode to the anode via the external circuit.

a) External Power Source: When a lithium-ion battery is connected to an external power source for charging, electrical energy is supplied to the battery.

b) Electron Flow: The external power source drives electrons from the cathode to the anode through the external circuit, creating a flow of electrons.

c) Lithium-Ion Movement: Inside the battery, lithium ions move from the cathode to the anode through the electrolyte, balancing the charge transfer caused by the electron flow.

d) Anode Intercalation: As lithium ions reach the anode, they are intercalated (inserted) into the anode material, typically graphite. This process stores energy in the battery.

B. Battery Discharging Cycle

During battery discharging cycle as shown in Fig 1 (b), lithium ions return to the cathode, while electrons move from the anode to the cathode via the external circuit, supplying power to the connected device.

a) External Load: During discharging, the battery is connected to an external load, such as an electronic device, which consumes the stored energy.

b) Electron Flow: Electrons flow from the anode to the cathode through the external circuit, providing power to the connected device.

c) Lithium-Ion Movement: Concurrently, lithium ions move from the anode to the cathode through the electrolyte, maintaining charge neutrality.

d) Cathode Intercalation: At the cathode, lithium ions are intercalated back into the cathode material, typically a lithium metal oxide. This process releases the stored energy, which is utilized by the external load.

The cyclical movement of lithium ions and electrons during charging and discharging allows the battery to store and release energy efficiently.

C. Remaining Useful Life (RUL)

The Remaining Useful Life (RUL) of a lithium-ion battery describes the estimated number of cycles or time the battery can continue to operate at a usable level before needing replacement. For applications such as electric cars, portable gadgets, and renewable energy storage systems, RUL is a critical metric for battery management systems (BMS) to ensure reliability and efficiency.

Factors Affecting RUL include:

State of Charge (SoC): This is the ratio of the available capacity (Q(t)) to the maximum possible charge that can be stored in a battery (Qn) as shown in Fig 1(c). SoC can be difficult to calculate precisely since there is no simple way to measure it directly.

Depth of Discharge (DoD): The extent to which the battery is discharged before recharging is known as DoD as shown in

Fig.1 (d). Shallow discharges tend to result in longer battery life compared to deep discharges.

Charging/Discharging Rate: Charge/discharge rates, expressed in terms of the "C-rate," measure the rate relative to the battery's maximum capacity. Different C-rates impact the charging time and battery longevity.

Types of Chargers: The charger's quality is crucial for the battery's longevity. Proper voltage and current regulation, overcharge protection, and temperature monitoring are essential for maintaining battery health.

Operating Temperature: High temperatures accelerate degradation and reduce capacity, while low temperatures increase internal resistance and cause capacity loss. Optimal temperature ranges (typically 20°C to 25°C) are crucial for maintaining performance and longevity.

III. MACHINE LEARNING METHOD FOR BATTERY CAPACITY (RUL) ESTIMATION

Estimating the remaining useful life (RUL) of lithium-ion batteries is crucial for maintaining the reliability and efficiency of battery-powered devices, such as electric vehicles and portable gadgets. Machine learning (ML) techniques have become increasingly important compared to traditional methods due to their ability to manage complex, nonlinear relationships in battery degradation data, resulting in more accurate RUL predictions. Predicting the RUL of these batteries is essential to optimize their usage and to plan maintenance or replacements.

A. Data Acquisition & Pre-Processing

The process starts with collecting data from battery management systems (BMS) that monitor variables such as voltage, current, temperature, and state of charge (SOC). This data provides a detailed history of the battery's performance, often covering multiple charge/discharge cycles. Pre-processing involves filtering to remove noise and outliers, normalizing to scale the data to a standard range, and extracting features that are relevant to battery degradation.

B. Feature Engineering

Feature engineering is critical for enhancing the accuracy of ML models. Relevant features for lithium-ion batteries might include temperature fluctuations, internal resistance, cycle count, depth of discharge (DoD), and charge/discharge rates. Advanced methods like signal processing and regression analysis can extract detailed information, such as voltage relaxation characteristics and charge capacity fading patterns. Combining domain expertise with these advanced techniques allows for the development of robust and accurate ML models.

C. Developing a Machine Learning Model for Battery Capacity (C)

The development of an ML model for predicting battery capacity involves several key steps as shown in Fig 2:

Selection of Applicable Battery Dataset: The initial step is to select a dataset with comprehensive and relevant information about the battery system under various operating conditions. The dataset should include data on electrochemical impedance spectroscopy, charge, and discharge cycles, reflecting real-world battery usage.

Classify Input Parameters: Identify the input parameters for predicting battery capacity. These should be measurable, accessible, and significant for capacity estimation. Common input variables include current, voltage, temperature, and time, with additional factors like battery impedance and charge transfer resistance considered depending on data availability.

Drawing Salient Features: Transform raw input variables into meaningful and practical features to enhance the ML algorithm's predictive capabilities. Techniques such as statistical analysis, signal processing, and dimensionality reduction can be used to extract features that capture the essential attributes and trends of battery behaviour.

Data Pre-Processing: Ensure the dataset is clean, reliable, and suitable for the ML algorithm through normalization, scaling, filtering, smoothing, outlier detection, and imputation of missing values. Effective pre-processing improves the accuracy and stability of the ML algorithm without introducing bias or distortion.

Data Splitting: Divide the dataset into training, validation, and test subsets. Training data is used to train the ML algorithm, validation data helps tune parameters and select the best model, and test data evaluates the final model performance. The splitting should be done randomly and proportionally to maintain data distribution consistency.

Selecting Machine Learning Algorithms: Choose appropriate ML algorithms capable of recognizing complex and nonlinear relationships between input features and battery capacity. Common algorithms for RUL prediction include Random Forest Regression (RFR), Extra Tree Regression (ETR), and K-Nearest Neighbors (KNN).

Model Creation: Construct the ML model using the selected algorithm(s), defining its structure, parameters, and hyper-parameters such as the number of layers, nodes, activation functions, learning rate, regularization methods, and loss functions. The goal is to optimize performance while avoiding overfitting or under fitting.

Model Training & Maturation: Train the model using training and validation data to identify patterns and relationships between input features and SOC values. This involves updating model parameters through optimization methods like gradient descent. The aim is to minimize the loss function and improve predictive accuracy, ensuring the model generalizes well to new data.

Model Testing & Validation: Evaluate the ML model's performance using test data, measuring metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) error. Validation involves assessing the model on a separate dataset to tune hyper-parameters and prevent overfitting, ensuring the model performs well in real-world applications.

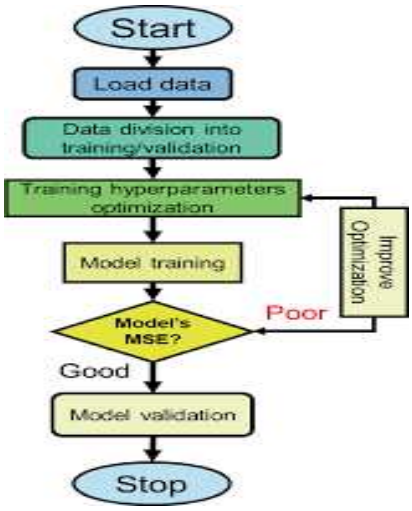


Fig. 2 Flow chart for Machine Learning model development

IV. NASA BATTERY DATASET [LI-ION 18650 (2AH)]

NASA Battery dataset for LIB 18650 (2Ah) was meticulously collected from a specially designed battery prognostics testbed developed at the NASA Ames Prognostics Center of Excellence (PCoE). The sophisticated experimental setup is designed to monitor and evaluate battery health and performance under various conditions. It enables precise control and tracking of multiple factors influencing battery performance, leveraging NASA PCoE's expertise in prognostics and health management (PHM).

The dataset includes a comprehensive range of parameters essential for understanding battery performance and health, such as voltage, current, temperature, and internal resistance. Three distinct operating profiles—charge, discharge, and electrochemical impedance spectroscopy—were applied to lithium-ion batteries at various temperatures. Discharges were conducted until the battery voltage reached predetermined thresholds, sometimes set below the OEM-recommended 2.7V to simulate deep discharge aging. Frequent charging and discharging cycles accelerated battery aging, with experiments continuing until the battery's end-of-life (EOL) criteria were met, resulting in a capacity drop from 2.0Ah to 1.4Ah.

Voltage Measurement: The voltage graph shown in Fig 3 (a) over time reveals that the initial voltage is 4.25 V, the voltage drops sharply as the battery transitions from a fully charged state. After the initial drop, the voltage decreases gradually, indicating stable discharge under a consistent load. Towards the end, a steep decline to below 2.75 volts signifies the battery nearing its end-of-life. A small uptick at the end suggests a recovery period due to reduced load or internal stabilization.

Overall, the graph depicts a typical lithium-ion battery discharge curve with phases of initial drop, stable discharge, and end-of-life decline, providing insights into battery performance and health.

Current Measurement: The current graph over time shown in Fig 3 (b) reveals a sharp drop at the start stabilizes around -1.0 amps, indicating consistent discharging. The current remains constant at approximately -1.0 amps for most of the period, signifying uniform discharge under a consistent load. Near the end, a sudden return to 0 amps suggests the discharge process has stopped, likely due to reaching a low voltage cutoff or manual test termination. Overall, the graph indicates a controlled and consistent discharge process, typical in battery testing scenarios where a fixed load is applied.

Temperature Measurement: The temperature graph in Fig 3 (c) shows that over time, temperature briefly dips to 5°C from 6°C before gradually increasing. The temperature shows a cyclical pattern of rising and falling, suggesting periodic heating and cooling phases due to charging and discharging cycles or varying load conditions. The overall trend is upward, peaking at approximately 12°C towards the end, indicating progressive heating during the test.Regular peaks and troughs indicate consistent periodic events causing heating followed by cooling phases, suggesting a controlled test environment with regular cycles.

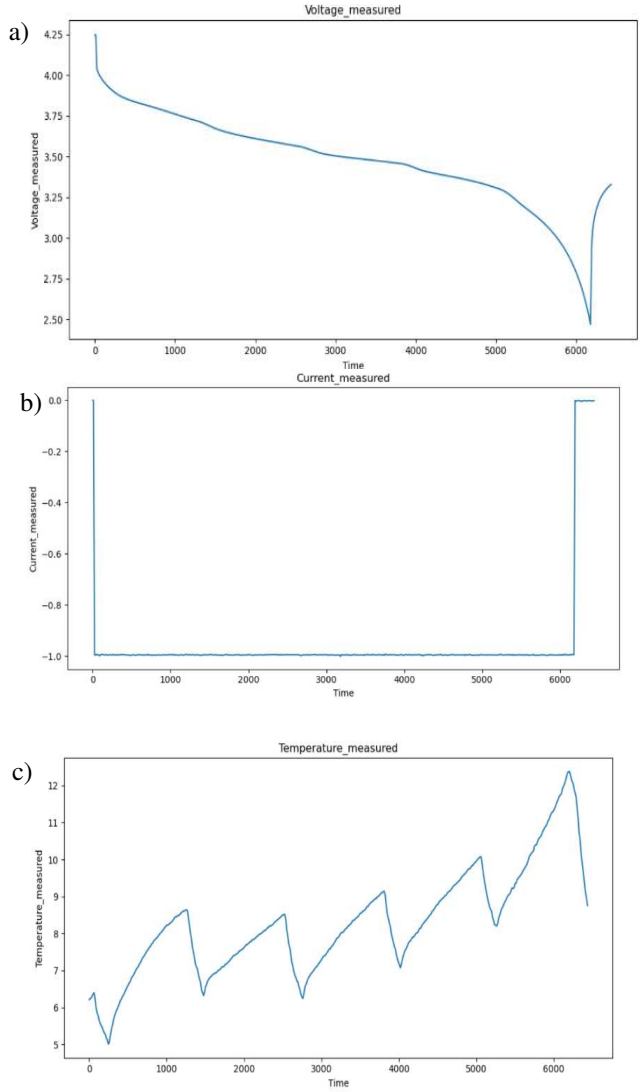


Fig. 3 a) Voltage Measured vs Time (s) b) Current Measured vs Time (s) c) Temperature measured vs Time (s)

V. MACHINE LEARNING MODELS

A. Extra Tree Regression

The Extra Trees Classifier shown in Fig 4, an ensemble learning technique, aggregates results from multiple decision trees to produce classification outcomes. Similar to the Random Forest Classifier, it differs primarily in its method of tree construction. In Extra Trees, each tree is built using the original training data, but during node splitting, each tree is given a random subset of features (denoted as k). The optimal feature is then selected based on criteria like the Gini Index. This randomness fosters the creation of diverse and uncorrelated trees. Feature selection is conducted by evaluating the normalized total reduction in the chosen splitting criterion for each feature during forest construction, resulting in the Gini Importance metric. Features are ranked by their Gini Importance, with the top k features being selected for use. During the training phase, each decision tree in the Extra Trees Forest is trained on the entire dataset, allowing the model to capture the intricate relationships between the predictors and the target variable, such as the Remaining Useful Life (RUL). For feature selection, the Extra Trees Classifier provides a straightforward yet effective approach by assessing the impact of each feature on the splitting criterion during tree construction, quantifying the importance of each predictor in forecasting RUL. This process helps practitioners prioritize features based on their

relevance, enhancing model interpretation and potentially improving predictive performance until a stopping criterion is satisfied, such as reaching the maximum depth or attaining the minimum number of samples per leaf [4].

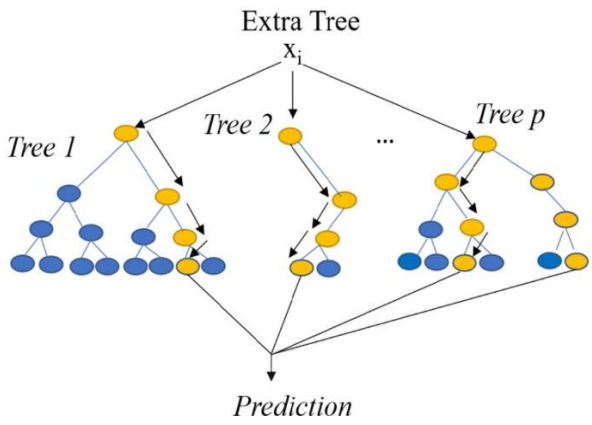


Fig. 4 – Extra Tree Regression (ETR)

B. Random Forest Regression

Random Forest Regression (RFR) shown in Fig 5 is a versatile machine learning technique adept at handling complex, high-dimensional data, even in nonlinear contexts. It excels in feature selection and importance ranking while leveraging the power of combining multiple decision trees to reduce overfitting and enhance generalization. In RFR, a dataset with six predictors (Voltage_measured, Current_measured, Temperature_measured, Current_load, Voltage_load, and Time) and one target variable (Battery Remaining capacity or RUL) is used. The goal is to train the RFR model to understand the intricate relationship between the predictors and the target, enabling accurate predictions of the battery's RUL with new input data.

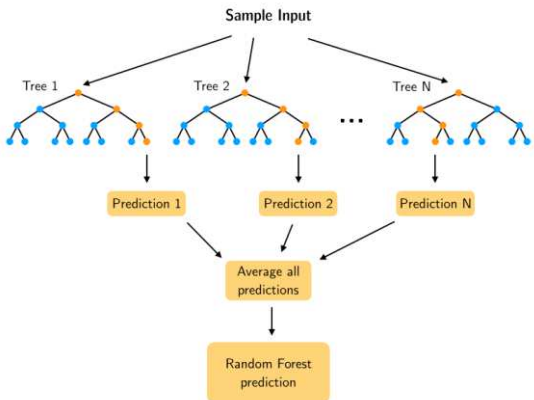


Fig. 5 – Random Forest Regressor Model

Training the RFR model involves specifying key parameters such as the number and size of trees, and the criteria for node splitting. The number of trees, a hyper parameter, affects the model's complexity and diversity. The dimensions of individual trees are controlled by parameters like maximum depth, minimum samples per leaf, and maximum features per split. To enhance randomness and reduce tree correlation, techniques like bootstrap sampling and feature bagging are used. Bootstrap sampling involves selecting random subsets of data for each tree, while feature bagging randomly chooses subsets of features for splitting nodes. These techniques strengthen the model's robustness and accuracy.

Training each tree involves recursively partitioning nodes until reaching predefined stopping criteria such as maximum depth or minimum samples per leaf. At each node, the optimal split among randomly selected features is determined using a criterion like mean squared error (MSE) or variance reduction. The aim is to maximize information gain or minimize node impurity, typically based on the MSE or variance of output values at the node. Through this iterative process, the RFR model is refined to effectively capture the intricate relationships within the data, enabling precise predictions of the battery's RUL under varying conditions.

C. K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a non-parametric, instance-based machine learning algorithm used for both classification and regression tasks (shown in Fig. 6). It predicts the label of a new data point by considering the majority vote (for classification) or the average (for regression) of its nearest neighbors in the feature space. KNN is based on the principle of similarity, assuming that similar data points should have similar labels or values. The algorithm determines the class or value of a new data point by analyzing the 'k' closest data points in the training set. The value of 'k' is a hyperparameter specified by the user, determining the number of neighbors to consider.

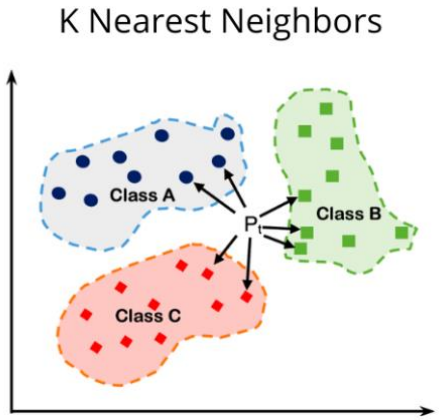


Fig. 6 – K-Nearest Neighbors (KNN) model

For classification, KNN identifies the 'k' nearest neighbors and takes a majority vote among them to classify the new data point. For regression, it computes the average (or weighted average) of the target values of the 'k' nearest neighbors to predict the value for the new data point. KNN is valued for its simplicity and ease of implementation, but its performance can be affected by the choice of 'k' and the distance metric used to measure similarity between data points. Common distance metrics include Euclidean distance, Manhattan distance, and cosine similarity.

However, KNN can be computationally expensive, especially with large datasets, as it requires calculating distances between the new data point and all training data points. To mitigate this, optimization techniques like KD-trees and ball trees can be used to speed up the search for nearest neighbors. Despite these challenges, KNN remains a useful tool in the machine learning toolkit, particularly when interpretability and simplicity are prioritized over predictive performance.

In the context of predicting battery Remaining Useful Life (RUL) or capacity using input parameters such as Voltage_measured, Current_measured, Temperature_measured, Current_load, Voltage_load, and Time from the NASA LIB battery aging dataset, KNN provides a straightforward yet powerful approach. Each battery observation in the dataset is represented as a point in a multi-dimensional space defined by its features. When a new battery observation is introduced, KNN calculates the

distances between this observation and all other data points in the dataset. The number of neighboring data points to consider, denoted as K, is determined by the user. Once the K nearest neighbors are identified, the algorithm proceeds with regression, computing the average RUL or capacity of the K nearest neighbors and assigning this value as the predicted RUL or capacity for the new observation. Selecting the value of K involves a trade-off: smaller values can lead to noisy predictions, while larger values may smooth out local fluctuations but can overlook important details. Thus, careful selection of K is crucial for optimal predictive performance. Additionally, feature scaling and normalization may be necessary to prevent features with larger scales from dominating the distance calculation.

IV. DEVELOPMENT OF MACHINE LEARNING MODEL FOR RUL PREDICTION

Development of machine learning models for predicting Remaining Useful Life (RUL) involves importing essential Python libraries such as numpy, pandas, matplotlib, seaborn, scikit-learn, and other required machine learning libraries for Random Forest Regression (RFR), Extra Trees Regression (ETR), and K-Nearest Neighbors (KNN) as shown in Fig 7(a). We then imported the metadata from the NASA dataset to understand the input and output features across various Excel files, focusing on the discharge dataset files crucial for our model development.To ensure data integrity and reliability, we conducted thorough data cleaning and pre-processing. This included examining each dataset for missing values and addressing them appropriately, as well as implementing outlier detection using the z-score test to eliminate data points with significant deviations from the mean. These steps were taken to prevent anomalies from negatively impacting the model's performance and accuracy. This meticulous data preparation laid a solid foundation for developing robust and accurate RUL prediction models.

A. Descriptive Statistics

We began our analysis by examining descriptive statistics shown in Fig 7 (b) to understand our dataset's characteristics. This thorough exploration provided valuable insights into its distribution and central tendencies, laying a strong foundation for subsequent model development. We prioritized features with strong relationships with Remaining Useful Life (RUL) to enhance prediction accuracy. Visualization techniques like correlation tables and heat maps helped to uncover patterns and identify variables significantly correlated with RUL. The heatmap shown in Fig 8 (a) depicted correlation coefficients between various parameters, indicating relationships through colors, with darker shades representing stronger correlations. These steps ensured our machine learning models focused on pertinent features, improving prediction accuracy and effectiveness.

Voltage measured exhibits a moderate positive correlation (0.37) with Current measured, suggesting that higher voltage is associated with increased current as shown in Fig 7. Conversely, it shows a moderate negative correlation (-0.36) with Current load, implying that higher voltage corresponds to lower current load. Similarly, Voltage measured demonstrates a moderate negative correlation (-0.31) with Time, indicating a decrease in voltage over time. Current measured displays a moderate positive correlation (0.51) with Current load, implying their simultaneous increase, and a moderate positive correlation (0.37) with Time, suggesting current rises during discharge. Temperature measured shows weak correlations with other parameters, except for a moderate positive correlation (0.21) with Voltage load, indicating higher temperatures are linked to higher voltage load. Current load exhibits a strong negative correlation (-0.75) with Capacity, indicating higher current load reduces battery capacity significantly. It also has a weak negative correlation (-0.17) with Voltage load and a weak positive correlation (0.20) with Time. Voltage load demonstrates a moderate positive correlation (0.41) with Capacity, indicating higher voltage load correlates with higher capacity. It also shows a weak negative correlation (-0.23) with Time, suggesting voltage load decreases slightly over time. Time displays a moderate negative correlation (-0.51) with Capacity, indicating capacity decreases with increasing discharge time, typical in battery discharge processes.

The density plot of Capacity values shown in Fig 7 (c) illustrates the distribution of capacity values, indicating multiple common values with a prominent peak around 2.0. The spread extends from about 0.5 to 2.5, revealing considerable variability. The line graph of capacity values against indices shown in Fig 7 (c) displays fluctuating trends with frequent peaks and troughs, indicating dynamic changes over the sequence. Statistical indices shown in Fig 7(b) summarize capacity values, with a mean of 1.4658 and a standard deviation of 0.3032, suggesting moderate spread and negative skewness (-0.7492). Kurtosis (0.2050) indicates slightly higher peaks and fatter tails than normal distribution. Following correlation analysis, we curated a subset of variables with robust relationships with RUL for machine learning model integration. This meticulous selection enhanced predictive capabilities. Min-max scaling on selected variables as shown in Fig 8 (b & c) facilitated data rescaling to a standardized range, promoting balanced model development. This technique mitigated the influence of outliers, bolstering model resilience against potential aberrations in the dataset.

a)

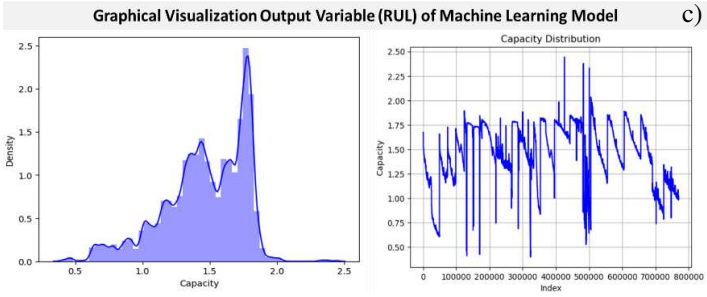
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn import metrics
import tensorflow as tf
import chardet
import os, tqdm
```

b)

Statistics for Capacity:

Statistical Indices	Value
Total Data Points	728484.000000
Mean	1.465766
Standard Deviation	0.303188
Skewness	-0.749151
Kurtosis	0.204983
Min	0.399484
25%	1.293464
50%	1.493520
75%	1.731884
Max	2.444062

c)



d)

```
#Descriptive statistics of NASA LIB Battery Discharge data set
Discharge.describe()
```

	Voltage_measured	Current_measured	Temperature_measured	Current_load	Voltage_load	Time
count	770070.000000	770070.000000	770070.000000	770070.000000	770070.000000	770070.000000
mean	3.417351	-1.578464	27.281331	1.487260	2.015373	1883.618060
std	0.419136	1.161964	14.062328	1.268142	1.134413	1390.592695
min	0.192738	-4.038796	2.904756	-2.000000	0.000000	0.000000
25%	3.282026	-2.010765	11.462661	0.998000	1.304000	795.047000
50%	3.473762	-1.989172	30.083127	1.998000	2.431000	1639.781000
75%	3.664314	-0.995509	36.023366	1.999000	2.776000	2620.781000
max	4.542427	0.030461	69.869746	4.148800	97.000000	6574.671000

Fig. 7 a) Importing required libraries in python b) Statistical indices for Output variable- Capacity c) Density plot for Output variable- Capacity (RUL) d) Descriptive statistics for all variables

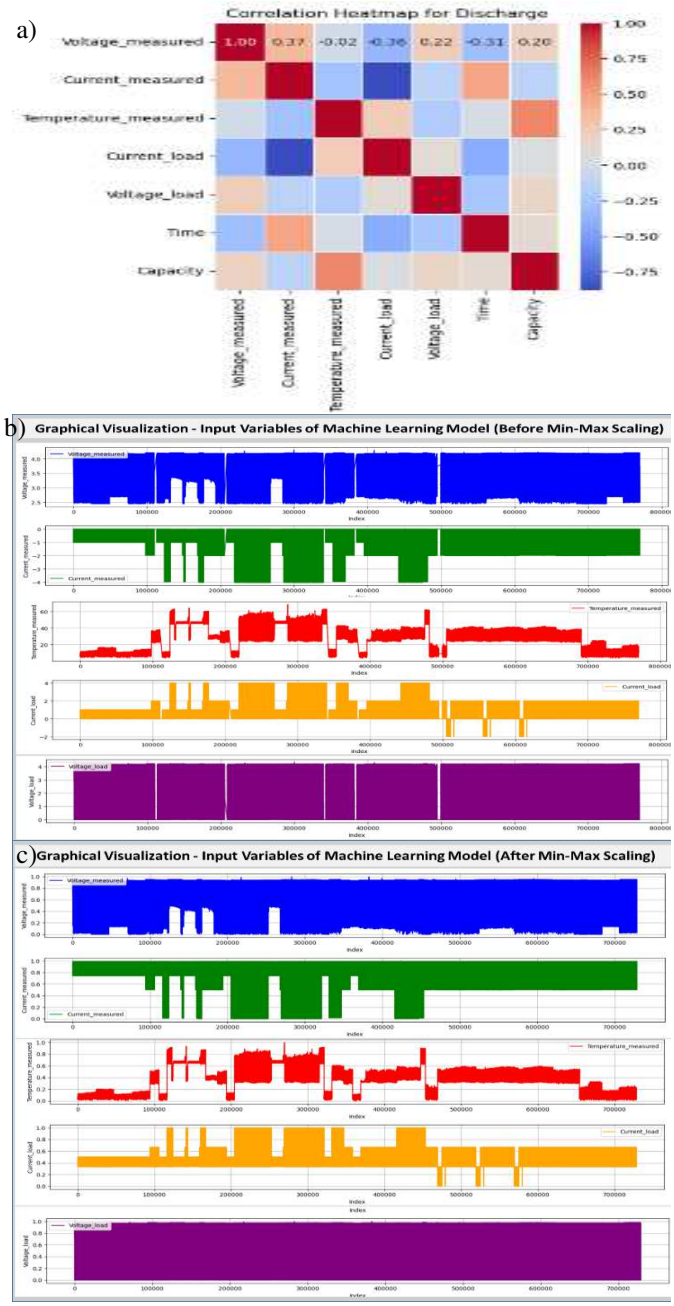


Fig.8 a) Heat map for Input & Output variables b) Input Variables before Min-max Scaling c)) Input Variables after Min-max Scaling

B. Model Development

This section delves into the process of selecting and building machine learning models to predict Remaining Useful Life (RUL) or Capacity, encompassing three main stages: data preparation, model selection, and model training as shown in Fig 9. Initially, the NASA LIB dataset is divided into input variables (Voltage_measured, Current_measured, Temperature_measured, Current_load, Voltage_load, and Time) and output variables (RUL or Capacity), with subsequent partitioning into training and testing sets in a randomized manner to ensure accurate evaluation of the model's performance. Regarding model selection, three algorithms are explored: Random Forest Regressor, Extra Tree Regressor, and K-Nearest Neighbor (KNN). Each algorithm requires tuning specific hyperparameters to achieve optimal performance. For Random Forest Regressor and Extra Tree Regressor, the critical parameter is the number of trees (n_estimators) in the ensemble. In this study, n_estimators is set to 25 and 42, respectively. This parameter manages the model's complexity and its ability to generalize to new data. While increasing the number of trees generally improves performance by reducing variance and enhancing prediction accuracy, there's a threshold where further increments may lead to overfitting, necessitating careful selection to balance accuracy and generalization. In KNN, used for regression tasks, the crucial hyperparameter is n_neighbors, determining the number of nearest neighbors considered for predictions. Here, n_neighbors is set to 5, striking a balance between capturing noise in the data (high variance) and oversimplifying patterns (high bias). This equilibrium ensures stable and accurate predictions while preventing both overfitting and underfitting, thereby enhancing the model's generalization ability.

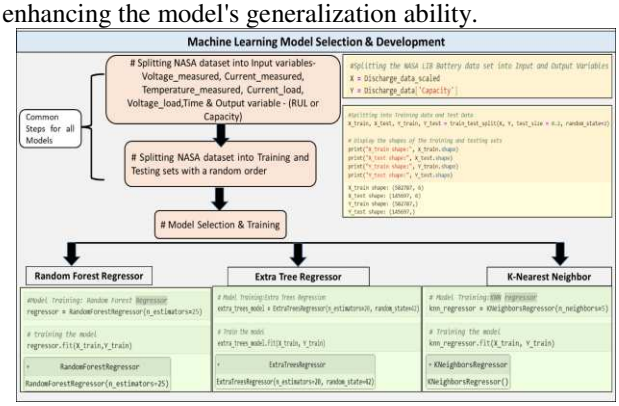


Fig.9 Machine Learning model selection

In this section, we delve into the process of developing machine learning models for SOC prediction (Fig. 8). The process basically entails three main steps: data preparation, model selection, and model training [3]. The first step involves splitting the Trip_AB dataset into input variables and output variables. The input variables include V, I, T & Ta, while the output variable is the SOC. The dataset is then split into training and testing sets with a random order [4]. This guarantees that the model is trained on a subset of the data and tested on unseen data to accurately measure its ability to generalize. For model selection, three ML algorithms are considered: RFR, GBR, and ANN each having its specific hyper parameters that need to be tuned for optimal performance. RFR and GBR both require specifying the number of trees (n_estimators) in the ensemble to control the complexity and generalization capability of the model [4]. In this study, n_estimators are set to 15 for RFR and 25 for GBR. Increasing the number of trees usually enhances the performance of the model, but there is a limit where it can start to have diminishing returns or over fit. The ANN model comprises multiple densely connected layers with specific activation functions and dropout regularization to prevent overfitting [1]. The first hidden layer has 128 neurons with a & the second has 64 neurons with ReLU activation, and a dropout layer with a rate of 0.3 follows to randomly drop 30% of neurons during training. The output layer has a single neuron to predict the SOC. The model is compiled utilizing the Adam optimizer with a learning rate of 0.0005 and employing MSE as the loss function to minimize the average squared differences between predicted and actual SOC values [10].The model is trained on the dataset for 500 epochs, with each epoch involving a complete pass through the data, split into batches of 64. A validation split of 20% is used to monitor performance and prevent overfitting.

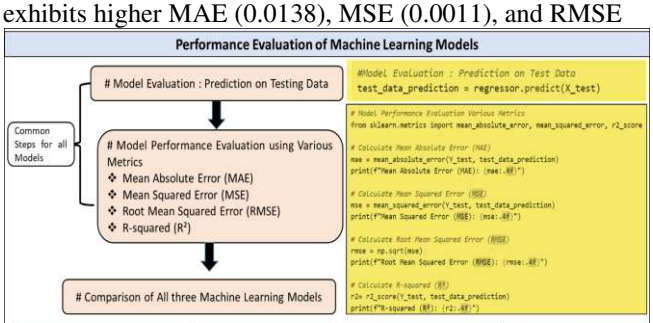
D.Model Evaluation

This section presents a comparative analysis of three machine learning models—Random Forest Regressor (RFR), Extra

Tree Regressor (ETR), and K-Nearest Neighbor (KNN)—using various performance metrics to reveal differences in their prediction accuracies as shown in Fig. 10. The metrics considered include Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2).

The Extra Tree Regressor (ETR) achieves the highest accuracy of 99.05%, consistently outperforming the other models across all metrics. It has the lowest MAE (0.0125), indicating the smallest average prediction error, and the lowest MSE (0.0009) and RMSE (0.0296), suggesting superior error minimization. Additionally, it boasts the highest R^2 value (0.9905), explaining the largest proportion of variance in the data, making it the most reliable model for predicting Remaining Useful Life (RUL) or Capacity.

The Random Forest Regressor (RFR) also performs well with an accuracy of 98.79%, albeit slightly behind the ETR. It exhibits higher MAE (0.0138), MSE (0.0011), and RMSE



Performance Evaluation Metrics	Random Forest Regressor	Extra Tree Regressor	K-Nearest Neighbor
Mean Absolute Error (MAE)	0.0138	0.0125	0.0161
Mean Squared Error (MSE)	0.0011	0.0009	0.0013
Root Mean Squared Error (RMSE)	0.0333	0.0296	0.0366
R-squared (R^2)	0.9879	0.9905	0.9854

Fig. 10 – Comparative analysis of all three machine learning models based on MAE,MSE, RMSE & R^2

(0.0333) compared to the ETR, indicating slightly larger prediction errors. However, its R^2 value of 0.9879 demonstrates significant variance capture, reflecting strong predictive capabilities. The K-Nearest Neighbor (KNN) model, with an accuracy of 98.54%, is effective but shows the highest errors among the three models, with MAE of 0.0161, MSE of 0.0013, and RMSE of 0.0366. Its R^2 value of 0.9854, although high, is the lowest among the models, indicating slightly less explained variance and marginally lower accuracy.

VI. RESULTS AND DISCUSSION

The performance of the three machine learning models—Random Forest Regressor (RFR), Extra Tree Regressor (ETR), and K-Nearest Neighbor (KNN)—was assessed for predicting battery Capacity or Remaining Useful Life (RUL) using the NASA LIB dataset. Various performance metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2), were utilized to evaluate the accuracy of each model. Scatter plots were also generated to visualize the relationship between the actual and predicted RUL values for each model. The scatter plots comparing actual versus predicted Capacity (RUL) values revealed distinct characteristics for each model.

A. Scatter Plot

The scatter plots shown in Fig 11 demonstrate that machine learning models can be powerful tools in predicting battery RUL or capacity. The accuracy of these predictions is vital & choosing the right model depends on the specific requirements of the application, such as dataset size, speed, and the need for real-time analysis. The visual comparison provided by these plots is an excellent way to quickly assess which model might be the best fit for a particular use case. The Random Forest Regressor (RFR) performance shows a high density of points along the diagonal line, indicating that the predictions are very close to the actual values. This suggests that RFR is highly accurate and reliable for this prediction task. RFR is known for handling non-linear relationships well, which is common in battery degradation data. Extra Tree Regressor (ETR) performance also exhibits a tight clustering of points around the diagonal, similar to RFR, which means it also provides accurate predictions. ETR can be particularly useful when we have a large dataset and need a model that is fast and less prone to overfitting. Its effectiveness in the scatter plot suggests it could be a good choice for real-time monitoring systems .K-Nearest Neighbor (KNN) while still showing clustering around the diagonal, has a broader spread of points. This indicates that KNN is somewhat less accurate than RFR and ETR but still within an acceptable range. KNN's simpler approach can be advantageous for smaller datasets or when a quick, rough estimate of RUL is needed.

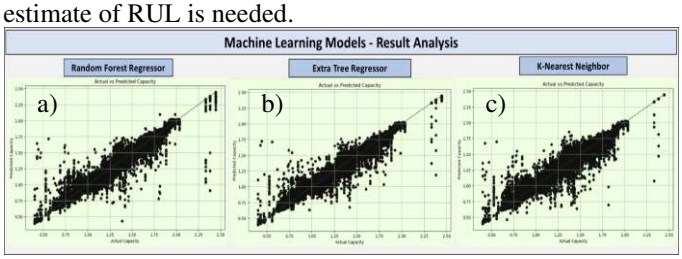


Fig. 11 Scatter plot between Actual & Predicted capacity a) Random Forest b) Extra Tree Regression c) K-Nearest Neighbor

B. Absolute Error Comparison

A comparative analysis of absolute error was conducted to evaluate the differences in model accuracy between the actual and predicted Remaining Useful Life (RUL) and capacity values derived from test data as shown in Fig 12. This analysis aimed to highlight the performance of three machine learning models: Random Forest Regressor (RFR), Extra Tree Regressor (ETR), and K-Nearest Neighbors (KNN). Randomly selected values from the test dataset indicated that all three models consistently demonstrated very low absolute errors. Among these, the Extra Tree Regressor (ETR) model exhibited the smallest range of absolute errors, spanning from 0.0004 to 0.0166. This narrow range suggests that the ETR model's predictions are very close to the actual values, indicating high accuracy. The ETR model's robustness and its capability to manage complex data structures likely contribute to this high level of precision. The Random Forest Regressor (RFR) model, while still performing well, showed a slightly larger range of absolute errors compared to the ETR, with values ranging from 0.0027 to 0.0200. Despite this, the RFR model's performance remains comparable to the ETR, suggesting that it is also a reliable model for predicting battery RUL or capacity. Similarly, the K-Nearest Neighbors (KNN) model exhibited absolute errors within the same range as the RFR, also between 0.0027 and 0.0200. This indicates that the predictive performance of KNN is on par with both the RFR and ETR models. These observations suggest that the ETR model provides the most accurate predictions of RUL, followed by the RFR and KNN models. The slightly higher absolute errors observed in the RFR and KNN models, when compared to the ETR, highlight the significance of careful model selection for State of Charge (SOC) prediction. The results underscore the effectiveness of the ETR model in

accurately predicting Lithium-Ion Battery (LIB) capacity or RUL, demonstrating its potential for practical, real-world applications.

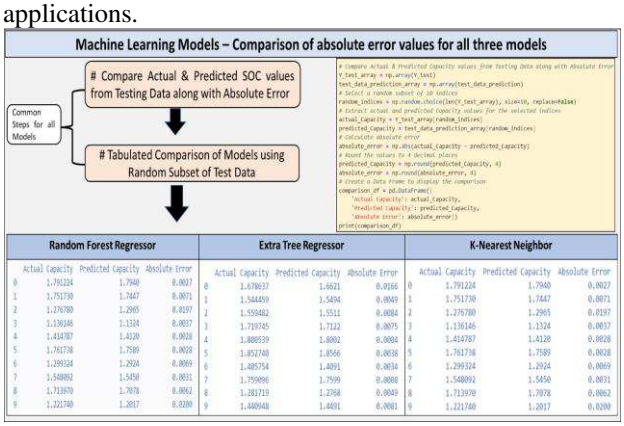


Fig. 12 Comparative analysis of all three model based on absolute error of predicted values

C. Feature Importance Plot

The feature importance plot provides insights into the relative contribution of each input variable towards predicting the target variable, in this case, the battery capacity or RUL as shown in Fig 13. The importance *scores* are scaled between 0 and 1, where a higher score indicates a greater impact of the variable on the model's predictions. The feature importance plot for the Random Forest Regressor (RFR) provides valuable insights into the predictive power of each feature when estimating battery capacity or Remaining Useful Life (RUL). In this plot, the feature 'Temperature measured' stands out as the most influential, with the highest importance score close to 0.5. This indicates that temperature measurements are critical in predicting battery behavior, likely because temperature can significantly affect battery chemistry and performance. The 'Time' feature is the second most important with importance score close to 0.15, suggesting that the duration of battery use or the age of the battery is a key factor in determining its RUL. This aligns with the understanding that batteries degrade over time and with usage. 'Current_load' and 'Voltage_measured' are also important features but to a lesser extent. These factors are essential in understanding the battery's operational conditions and how they might accelerate or decelerate the degradation process. Conversely, 'Voltage_load' and 'Current_measured' have the least importance scores, implying that while they do contribute to the model, their impact on the prediction is not as strong as the other features. This could be due to these features not varying as much as temperature or time, or possibly because they are correlated with other more dominant features. Similarly, the feature importance plot for the ETR also ranks the feature 'Temperature_measured' as the most influential, with the highest importance score close to 0.5. The second most important feature is 'Time' with importance score close to 0.15. It represents the operational duration of the battery. Longer usage accumulates wear and tear, leading to capacity degradation. Batteries age over time, and this feature captures that effect. In this case also 'Voltage_load' and 'Current_measured' have the least importance scores

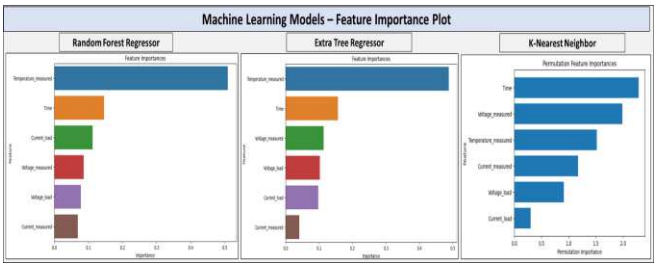


Fig 13 Feature Importance plot

The permutation feature importance plot is a valuable tool for understanding which features are most influential in the predictions made by a K-Nearest Neighbor (KNN) machine learning model for battery capacity or Remaining Useful Life (RUL). This method involves shuffling the values of each feature individually and measuring the impact on the model's prediction accuracy. The rationale behind this approach is that if shuffling a feature significantly deteriorates the model's performance, that feature is considered important.

Features that contribute significantly to accurate predictions will show a marked decrease in model performance when their values are permuted. Conversely, features that have little to no impact on the model's accuracy when shuffled are deemed less important. In this case 'Time' feature comes out as the most influential input variable with importance score of 2.5 followed by time with a feature importance score of 2.0. 'Temperature_measured' has an importance score of 1.5 followed by 'Current_measured', 'Voltage_load' & 'Current_load'. These insights can be helpful in optimizing battery performance and enhance RUL predictions.

VII. CONCLUSION

This study investigated the application of machine learning techniques in predicting the Remaining Useful Life (RUL) or capacity of batteries, using the NASA Lithium-Ion Battery (LIB) dataset to understand the influence of various parameters on battery life cycles. Three machine learning models—Random Forest Regressor (RFR), Extra Tree Regressor (ETR), and K-Nearest Neighbor (KNN)—were evaluated based on metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2). Results indicate that the Extra Tree Regressor (ETR) model was the most accurate for predicting battery capacity (RUL), exhibiting the lowest prediction errors and strong linear correlation with actual values. 'Temperature_measured' and 'Time' emerged as critical variables affecting RUL prediction, while other parameters had lesser influence. Recommendations for further exploration and implementation include enhancing model performance through hyper-parameter optimization and ensemble methods, exploring new feature engineering techniques, improving data pre-processing methods, integrating real-time data, refining model evaluation and validation processes, deploying models in practical applications, exploring advanced algorithms, collaborating with domain experts, and extending research to diverse operating conditions and battery types.

REFERENCES

[1] Zhu, Jun, et al. "RUL prediction of lithium-ion battery based on improved DGWO-ELM method in a random discharge rates environment." Ieee Access 7 (2019): 125176-125187.

[2] Qu, Jiantao, et al. "A neural-network-based method for RUL prediction and SOH monitoring of lithium-ion battery." IEEE access 7 (2019): 87178-87191.

[3] Ren, Lei, et al. "Remaining useful life prediction for lithium-ion battery: A deep learning approach." Ieee Access 6 (2018): 50587-50598.

[4] Liu, Jian, and Ziqiang Chen. "Remaining useful life prediction of lithium-ion batteries based on health indicator and Gaussian process regression model." Ieee Access 7 (2019): 39474-39484.

[5] Catelani, Marcantonio, et al. "Remaining useful life estimation for prognostics of lithium-ion batteries based on recurrent neural network." IEEE Transactions on Instrumentation and Measurement 70 (2021): 1-11.

[6] Ma, Yanying, et al. "The capacity estimation and cycle life prediction of lithium-ion batteries using a new broad extreme learning machine approach." Journal of Power Sources 476 (2020): 228581.

[7] Lee, Chul-Jun, et al. "Real-time prediction of capacity fade and remaining useful life of lithium-ion batteries based on charge/discharge characteristics." Electronics 10.7 (2021): 846.

[8] Li, Penghua, et al. "State-of-health estimation and remaining useful life prediction for the lithium-ion battery based on a variant long short term memory neural network." Journal of power sources 459 (2020): 228069.

[9] Zhao, Jingyuan, et al. "Machine learning for predicting battery capacity for electric vehicles." ETransportation 15 (2023): 100214.

[10] Hasib, Shahid A., et al. "A comprehensive review of available battery datasets, RUL prediction approaches, and advanced battery management." Ieee Access 9 (2021): 86166-86193.